

A History of Modern Philosophy

Introduction and Tenth Book

Harald Høffding

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Translator's note. This volume contains two parts of Høffding's two-volume history: the Introduction, in which he sets out his conception of philosophy and of what a history of philosophy is for, and the Tenth Book, on the philosophy of the second half of the nineteenth century. The remainder of the work is available in English in B. E. Meyer's translation of 1900, made from the German edition. The Introduction translated here is later than Meyer's text and refers to a work of 1902; the Tenth Book is given because its subject — Robert Mayer, the conservation of energy, and the idealist and critical reactions to materialism — is the background to arguments translated elsewhere in this edition.

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Introduction

It will be natural that, when a new presentation of an important period in the history of philosophy appears, the question should be asked what conception of philosophy the author takes as his ground, and what significance and value he ascribes to the history of philosophy. My answer is that the studies whose provisional conclusion is marked by the publication of this book had precisely the aim of orienting me with respect to what philosophy really is. As we come to know a person through his biography, so we must be able to come to know a science through its history. And it will be all the more natural to try this road, since experience constantly shows that in the domain of philosophy conflicting ways of seeing assert themselves, so that there is no presentation of philosophy to which one could point as giving an exhaustive concept of it. It will then be natural here — as in the domain of religion — to apply the comparative method. The history of philosophy treats the attempts which individual thinkers have made to discuss the ultimate problems of knowledge and of life. The most important result will be to get clear about which problems philosophy occupies itself with, how those problems are posed at different times, and by what the different ways of posing them, and of attempting to solve them, are determined in detail. If such a result can be approximately reached, the study of the history of philosophy will be of no slight importance for continued philosophizing.

For my own part I have been strengthened, by the study of the history of modern philosophy, in the assumption I had arrived at by another road: that philosophical inquiry turns upon four principal problems. These four problems are now to be briefly characterized. A more searching discussion and comparison of them I have given in the treatise *Filosofiske Problemer* (University Programme 1902).

1. The problem of knowledge (the logical problem). However different the several sciences may be in respect of subject and of method, they all nevertheless work with human thinking. Every time they form a concept, utter a judgment or draw an inference, they presuppose the general forms and principles of thinking. Here, then, there shows

itself the possibility of a peculiar discipline, one which investigates the forms in which thinking moves and the principles it must by its nature take as its ground, whatever the objects it may concern. This discipline, *formal logic*, treats however only one part of the problem of knowledge. Those forms and principles do not lead beyond thinking itself, but only make it possible for thinking to be in agreement with itself, to be consistent. Every time they are applied to given phenomena, which our thinking has not itself constructed but must take as they come, the question arises with what right that application takes place — with what right we assume that not only our thinking, but existence itself, which expresses itself in the given phenomena, is consistent, agrees with itself. Herewith shows itself the possibility of a discipline which investigates the conditions of a knowledge of existence, and the limits of such knowledge. This discipline is *epistemology*.

2. The problem of existence (the cosmological problem) arises when it is asked how the existence of which we are members must be supposed to be constituted, when we draw the ultimate consequences of all that we know, or can conjecture with the help of the most probable hypotheses. We call this problem the cosmological (from *kosmos*, world, and *logos*, doctrine), because it leads to a discussion of the possibilities that present themselves to thought when it seeks to work up what is given in experience into a general world-view, or when by speculation's bold road it attempts to construct one. The different philosophical systems are attempts in this direction — attempts whose value depends on how significant and how comprehensive are the experiences they take as their ground, and how much consistency and power of combination is displayed in their constructions.

3. The problem of valuation (the ethical-religious problem) arises through this, that we do not merely stand to existence in an intuiting and understanding relation, but also have our feeling set in motion by it, so that we utter judgments which ascribe value to it or deny it value. Of particular importance are the judgments we utter upon human actions, our own and others'. All such appraisal rests — like all knowledge and understanding — upon certain presuppositions, which it is a matter of exhibiting and determining. This task belongs under the ethical problem. When the valuation concerns not merely human actions and institutions but existence, life as a whole, the religious problem arises, which leads especially to a discussion of the relation between the ethical ideals and actual existence, so that it becomes a combination of the cosmological and the ethical problem.

4. The problem of consciousness (the psychological problem). For all three of the problems indicated it shows itself that their treatment presupposes an empirical

acquaintance with human conscious life. Psychology presents the *actual* development of human knowledge, which must be known before one can discuss the conditions of knowledge's *validity*. And since the relation between the spiritual and the material is a principal point within the problem of existence, psychology is presupposed in cosmology as well. As regards the problem of valuation, finally, psychology discusses partly the constitution of the very feelings that lead us to make valuations, partly the possibilities which actual conscious life presents for a further development in the direction valuation demands. In consequence of this close relation to the philosophical problems, psychology is itself to be reckoned a part of philosophy; and one will therefore also, conversely, quite naturally come in psychology to touch upon philosophical problems. And even if one were to declare all three of the preceding problems insoluble, or arisen through misunderstanding, the psychological problem — the question of the nature and the laws of conscious life — would still remain as philosophy's last stronghold.

— If it is asked what factors will, in the nature of the case, come to influence the treatment and the answering of these problems, the first to be named must be the personality of the one who philosophizes. The problems named have this in common, that they lie at the boundary of our knowledge, where the exact methods fail; it cannot then be otherwise than that the inquirer's personality must come to determine his train of thought, without his needing to be conscious of it. The "personal equation" must be of greater importance in philosophy than in other scientific domains. The historical, comparative method therefore becomes of particular importance here; by its help the personal element can most easily be exhibited. That personal element is not always something to be pushed aside; its presence is often a condition of a problem's being able to come to be at all. There are thoughts which can come to be only upon a determinate psychological soil. — It will, in the second place, depend upon which observations are taken as the ground. Here the development of natural science acquires especially great importance for modern philosophy. As will appear, the decisive problems of modern philosophy are determined by the very fact that modern natural science came into being. But to this there come as well — especially as regards the problem of valuation — historical conditions, intellectual movements in other domains. — Finally, the posing of problems and their solution are determined by the consistency with which the starting points once seized are held fast and carried through.

Both the characterization and the criticism of every philosophical attempt will always lead back to these three points. Among them I lay most weight, in the presentation that follows, upon the first two. An inconsistency in a great thinker is often precisely a

natural consequence of *several* trains of thought having opened themselves to his genius, without his having been able to pursue them far enough to discover their mutual conflict. And it may be of the greatest importance that these different trains of thought be brought out. Best of all, of course, is when profundity and consistency are united.

An attempt at posing or at solving a problem may have a twofold interest. In the first place it may be regarded as a symptom, as the historical expression of intellectual currents. From this side the history of philosophy is a part of general cultural history. In the second place it may be examined with respect to the real, definitive clarity that has been reached by it. This consideration is more specifically philosophical and technical. The two sides from which the philosophical phenomena can be regarded will of course, in the case of different phenomena, stand in very different relations to one another. Sometimes the cultural-historical interest may even stand in inverse relation to the purely philosophical.

If I am briefly to indicate wherein my presentation of the history of modern philosophy differs from its predecessor in our literature — Brøchner's work of 1873–74 — I would name first the particular weight I lay upon the personal factor, upon the relation to empirical science, and upon the cultural-historical significance of the philosophical phenomena. Next I would point to my endeavour to give the posing of problems precedence over their solution. Solutions may die, and the problems live on; otherwise philosophy would not have had so long a life as it has had. Finally it goes without saying that a renewed study of the sources, as well as the use of the abundant literature of recent years, has set many phenomena in a new light. But in spite of all this it has been a dear thought to me, in this work, to be able to continue Brøchner's labour in the field of the history of philosophy; and I can only hope that my book may not be unworthy to be consecrated to the memory of that noble inquirer, whose admiration for the heroes of thought was at once so deep and so understanding.

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Tenth Book

PHILOSOPHY IN GERMANY 1850–1880.

Of the two great tendencies in nineteenth-century philosophy it was positivism that had best maintained its connection both with the thought of the eighteenth century and with empirical science. The philosophy of Romanticism, by contrast, was a definite and conscious attempt at reaction against both these tendencies; it wanted even to do over again what had been founded with the rise of natural science in the seventeenth century. In Germany, the hearth of Romanticism and of its philosophy, that tendency was the dominant one at the middle of the century. The adherents of the critical undercurrent were the only ones who held fast to continuity with the other sciences — leaving aside Schopenhauer and Feuerbach, who at that time still stood as solitary thinkers for whom the hour of recognition had not come.

As little as positivism was to be explained as a reaction against Romanticism, so little is the philosophical movement which takes place in Germany after the middle of the century to be explained as a continuation of Franco-English positivism. It has its presuppositions in Germany itself. It receives its problems above all from the renewed rise of natural science towards the middle of the century. Not only do natural-scientific studies and results now become the object of more general interest than in the earlier part of the century, when poetry, religion and speculative philosophy had laid claim to men's minds; natural science itself now returned, with clearer consciousness, to the great principles its founders had in their day set up. The demand for an understanding of outer nature as a series of demonstrable causes and effects — in other words, for a mechanical explanation of nature — was now asserted again, as in the seventeenth century. And at the same time a new and comprehensive law of nature was discovered and demonstrated: the law that in physical nature no energy arises or perishes, but that what happens in every apparent arising or perishing is a conversion of energy into another form, and in such a way that the different forms of energy stand in a determinate quantitative relation to one another. This law — which, together with Darwin's doctrine of the origin of species

through the struggle for existence, is the most significant result of natural-scientific investigation in the nineteenth century — could not but set philosophical thinking in motion, just as in their day the Copernican theory and Galilei's founding of mechanics had led philosophy into new paths. For German philosophy in particular it had to become a principal question: how far can the thoughts developed by the philosophy of Romanticism be held fast, if one accepts the new natural-scientific way of regarding things? This question was answered in three different ways. Modern materialism rejected those thoughts as pure illusions and proclaimed the doctrine that only the material exists, as a necessary consequence of natural science. Lotze and Fechner, by contrast, attempted to exhibit the fundamental thought of speculative philosophy of religion as the last and concluding presupposition of the picture of existence that can be built up by the help of the natural-scientific method. Finally Albert Lange and Eugen Dühring approached critical philosophy and positivism, by bringing out the significance of the problem of knowledge and by maintaining practical idealism's independence of empirical science, while at the same time insisting upon experience's right to determine the actual content of the world-picture.

1. Robert Mayer and the Principle of the Conservation of Energy

Not only in philosophy, but in the other sciences too, knowledge moves forward in rhythmic fashion, through actions and reactions. At the close of the eighteenth century natural science had a great and fruitful period, in which some of the most important truths in the domains of chemistry and physiology were discovered. LAVOISIER introduced the quantitative method into chemistry, and by its help the truth of the old thought was established that no matter arises or perishes, but that the same mass of matter subsists through all changes, under different forms. Chemistry was founded as an exact science. PRIESTLEY, INGENHOUS, SENEBIER and SAUSSURE found the principal laws of the metabolism of plants and animals, and thereby founded the great doctrine of the circulation of matter in nature, whereby the organic and the inorganic world appear in their close mutual connection. It was discovered that it is under the influence of sunlight that organic matter is formed in the green plant cells by the taking up and dissolution of the carbonic acid of the air. The carbon stored in the plant cells then serves animals for nutriment; by the animal functions a combustion takes place, and the carbonic acid thereby formed passes out into the air in respiration, whence the same circulation can

begin anew. A great cosmic connection had here been exhibited.¹ The picture of the world sketched by Copernicus, Bruno, Kepler and Galilei, which Newton had enriched by exhibiting universal attraction as a binding and ordering tie, hereby received a new and significant enrichment.

But the age was too much taken up with revolution and world war, with Romanticism and speculation, and soon with orthodoxy and mysticism as well, for these great ideas to gain the influence upon general opinion that they deserved. Even within natural science their victory was hindered by other tendencies. It was within the science of organic life that resistance was chiefly offered to the new points of view, because men were still governed by the reaction against the Cartesian doctrine of the organism as a machine. The expressions peculiar to the organism were derived from a peculiar vital force, supposed to be wholly different from the other forces of nature. This mode of explanation, so-called vitalism, could not but lead to a rejection of the attempts to draw the organic phenomena into the universal circulation of matter.² Besides, the prevailing interest among botanists and zoologists was to describe the forms and determine their systematic place, not to find the developmental process through which the forms come to be, and the causes that determine that process. Romantic philosophy of nature favoured this aesthetic and formal conception — and had itself proceeded from it. When “philosophy of nature” is often given the whole blame for the fact that the points of view of the close of the eighteenth century did not sooner prevail in botany and zoology, it must not be forgotten that it was quite as much an effect of the condition of natural science as a cause of it.

From about 1840 a change came about in this respect. Life began to be conceived as something other than a play of forms or a revelation of ideas. DUMAS and LIEBIG insisted upon the significance of the chemical processes for plant and animal life. And from the medical side too a strict mechanical explanation began to be demanded of the processes that go on in the organism. In some of his earliest writings HERMANN LOTZE worked in this direction, especially in *Allgemeine Pathologie und Therapie als mechanische Naturwissenschaften* (1842) and in the treatise *Leben, Lebenskraft* (1843, in R. Wagner’s *Handwörterbuch der Physiologie*). The task was set of making physiology a purely mechanical science. Most epoch-making, however, was the fundamental proposition set up by the physician and physicist ROBERT MAYER (b. 1814, d. 1878) in his work *Die*

¹On “the period of the great discoveries” at the close of the eighteenth century see Rasmus Pedersen: *Planternes Næringsstoffer*. Historisk Indledning. København 1883, p. 44 ff.

²On vitalism in its various forms see Louis Peisse: *La Médecine et les Médecins*. Paris 1857. I, p. 226–297. — It must be firmly held that by vitalism is here meant a tendency which finds the *explanation* of vital phenomena in the assumption of a “vital force” — not a tendency which maintains that there is much in the vital phenomena that has hitherto not been explained by physical and chemical causes. See my treatise *Om Vitalisme* (in “Mindre Arbejder”).

organische Bewegung in ihrem Zusammenhange mit dem Stoffwechsel (1845): that in the process of life there takes place only a transformation of force as well as of matter, never a creation of them. Mayer here points back to the great law he had discovered a few years earlier, when he set the law of the conservation of force beside the law of the conservation of matter deriving from Lavoisier.

Robert Mayer's life as an investigator is borne by a single thought, which he came upon early; but it was the tragedy of his life that he was unable to carry it through in detail by experimental means. As a child he had tried to construct a *perpetuum mobile*, and it made a strong impression upon him that he did not succeed. From that time he brooded constantly upon the relation between cause and effect — first and foremost within physiology, which as a medical man lay nearest to him, but thereafter within chemistry and physics as well. On a voyage to the East Indies as ship's doctor he arrived at the new idea of the imperishability of force in nature, under the influence partly of investigations into the origin of organic heat, partly of the fact that heat arises in the wave-motion of the sea. His point of departure is thus definitely observation. But that observation led him to ask whether a law similar to the one in chemistry might not hold in physics too, so that not only matter but force also was imperishable, and every change a mere conversion. And he then came back to his earlier broodings upon cause and effect. It was for him a self-evident proposition that there could not be more in the effect than in the cause: *causa æquat effectum*! It follows from "the law of the logical ground"! Mayer thus does not distinguish, any more than the dogmatic philosophers do, between ground and cause; thereby he comes, like William Hamilton and Herbert Spencer, to regard it as a matter of course that the cause cannot become nothing when the effect arises, but that the effect must furnish an equivalent for what seemed to vanish with the cause. Hitherto men had rested content with the fact that a motion ceased when it met a sufficiently strong resistance, and that heat arose from frictional resistance. But — Mayer asked — does the motion then become nothing, and does heat then arise out of nothing? If that is so, then "the red thread of science" is cut through, just as it would be if in chemistry we were to assume that oxygen and hydrogen became nothing when they combined, and that water then came to be out of nothing. Just as we assume that oxygen and hydrogen are transformed into water, so we must also assume that the motion does not become nothing, but is transformed into heat.³ And by the expression *transformed* Mayer meant

³Cf. Mayer's letter to Griesinger of 20 July 1844: "Wenn ich sage, Wärme lässt sich in Bewegung verwandeln, und umgekehrt, so will dies nichts meinen, als dass zwischen Wärme und Bewegung finden hin und her dieselben quantitativen Beziehungen statt." (*Kleine Schriften und Briefe von Robert Mayer*. Herausg. von Weyrauch. Stuttgart 1893, p. 225.) — In his paper *Robert Mayers Entdeckung und Beweis des Energieprinzips* (in the Festschrift for Sigwart. Tübingen 1900) A. Riehl illuminates Mayer's doctrine from

to designate only that a constant quantitative relation subsists between the cause, which disappears, and the effect, which comes in its place. If the train of thought put forward is correct, experience must establish such a relation of equivalence between the different forms of force that succeed one another. On the basis of experiments already available, Mayer sought already in his first paper (*Bemerkungen über die Kräfte der unbelebten Natur*, printed in Liebig's "Annalen der Chemie und Pharmacie", 1842) to determine the quantitative relation between heat and motion. It was this idea of an invariable relation of magnitude that was the main thing for Mayer, and that in the following period proved so fruitful for natural science. It was a significant extension of the abstract law of the conservation of force, already maintained by Huyghens and Leibniz, so as to hold for the relation of the different forces of nature to one another. Mayer derived from his train of thought that there really exists only a single force, which appears under different forms, and these stand to one another in a determinate quantitative relation.

Although Mayer's attempt to give a philosophical derivation of the law of the conservation of energy does not hold, his train of thought has nevertheless an epistemological interest. For it was in itself fully legitimate to raise the question whether between the event we call the cause and the event we call the effect there might not obtain a relation similar to that between logical ground and logical consequent. And Mayer's train of thought leads naturally to raising that question. Experience must then decide what the answer is to be. — It is interesting that the Danish physicist COLDING, who in the year after Mayer's first paper was printed set up the proposition of the conservation of energy by a road of his own and confirmed it by experiment, proceeded from the view that it was really a proposition of reason; and likewise that HELMHOLTZ, in his paper *Ueber die Erhaltung der Kraft* (1847), proceeded from an epistemological postulate. The Englishman JOULE, who like Colding and Helmholtz had come by a road of his own to the same result as Mayer — a remarkable example of how several may simultaneously be on the track of one and the same discovery — proceeds predominantly experimentally, but nevertheless indicates the a priori improbability that force should be annihilated without a corresponding effect.⁴

an epistemological point of view.

⁴"If the resistance to electrolysis which is over and above that due to chemical change were not accounted for elsewhere, it would prove the annihilation of a part of the power of the circuit, without any corresponding effect. We shall see that this is not the case, but that in the evolution of heat, where the excess of resistance takes place, an exact equivalent is restored" (Joule: *On the Heat evolved during the Electrolysis of Water*. 1843. Scientific Papers. I. London 1884, p. 115). "We might reason, a priori, that absolute destruction of living force cannot possibly take place, because it is manifestly absurd to suppose that the powers with which God has endowed matter can be destroyed ... but we are not left with this argument alone, decisive as it must be to every unprejudiced mind" (Joule: *Matter, Living Force and Heat*. 1847. Scientif. Pap. I, p. 269). — Joule's standpoint is therefore not in principle different from Mayer's and

The new proposition had difficulty in winning recognition, but gradually made its way in the course of eighteen years (1842–1860), especially as it proved fruitful in leading to new investigations and discoveries. Its great importance consists precisely in the determinate questions it leads the investigator of nature to ask, every time a peculiar manifestation of force appears or ceases. Just as it is the importance of the general causal proposition that at every change which occurs it leads us to look about for a preceding change of which the new one might be the effect, so it is the importance of the proposition of the conservation of force — or, as one now prefers to say, following the proposal of English investigators, of energy — that one is at once brought to set oneself the task of finding out in what relation the manifestations of force that succeed one another stand to one another.

Philosophically regarded, the question had above all to become how the spiritual phenomena stood to the new law. It is characteristic in this respect that almost all its discoverers proceeded from decidedly spiritualistic and teleological views. Mayer repeatedly declared himself a definite opponent of materialism, and expressed as his conviction that the scientific truths stand to the Christian religion as brooks and rivers to the ocean. At the congress of natural scientists in Innsbruck in 1869 he made a declaration to this effect, to the offence of Carl Vogt and his friends. Colding's standpoint is seen from the following utterance: "My first thought that the forces of nature must be imperishable I drew from the view that the forces in nature must be akin to the spiritual in nature, to the eternal reason as well as to the human mind. It was thus the religious conception of life that led me to the thought of the imperishability of the forces of nature." Colding and Joule conceive that God at the creation laid a certain sum of force into nature, and that this sum can neither be increased nor diminished as a whole, but only distributed in different ways — as Descartes in his day assumed of motion. And both assumed that with the conservation of energy the conservation of what is valuable in the world was secured, since they no more than Leibniz distinguished between energy itself and its employment in favour of life and development.⁵

Colding's. For the latter compare an utterance in his *Undersøgelse om de almindelige Naturkræfter og deres gensidige Afhængighed*. (Videnskabernes Selskabs Skrifter. Femte Række. Naturv. mathem. Afd. II, p. 129): "The thought that an activity could disappear in the corporeal without appearing again as operative cause seems to me contrary to reason." — It is of interest in this connection to recall that Lavoisier too proceeded in his day from a fundamental proposition he regarded as a matter of course, namely that the weight of a compound body is equal to the weight of the substances of which it consists. Self-evident as that presupposition was, it nevertheless had to be proved to those who took heat for a substance; for if that assumption were correct, the release of heat which accompanies a chemical combination would have to entail a diminution of weight. Compare Ernst von Meyer: *Geschichte der Chemie*. Leipzig 1889, p. 137.

⁵Mayer: *Kleinere Schriften und Briefe*, p. 339 f., 460. *Die Mechanik der Wärme*. 3. Ausg., p. 356 f. — Colding: *Naturvidenskabelige Betragtninger over Slægtskabet mellem det aandelige Livs Virksomheder og de*

But a closer investigation and grounding of these presuppositions the natural scientists named do not undertake; they proceeded, like the philosophers of the seventeenth century, from given religious representations. A closer investigation of the relation of the law exhibited — and of the knowledge of nature so far won generally — to the phenomena of spiritual life had therefore naturally to be called forth. And it is then no wonder that the ways parted, and that different tendencies stepped forth against one another. All that was given was this: that when one kind of physical energy ceases, another kind of physical energy enters, in a certain determinate quantity, in its place. The question was what significance this knowledge must consistently come to have for the conception of the world.

2. Materialism

The strongest reaction against the philosophy of Romanticism — apart from crass orthodoxy — is marked by the materialist literature which flourished in Germany at the middle of the century. If there the Idea had been everything, now matter became everything. Enthusiasm for natural science, and for the great points of view it had succeeded in carrying through in the doctrine of the conservation of matter and of force, could naturally lead to regarding that doctrine itself as a sufficient philosophy, and to explaining every side of existence by its help. The simplest way of solving the question what significance the law of the conservation of matter and force has for our world-view would undeniably be to say that this law contains our whole world-view. The philosophy which modern materialism sets up will be nothing other than the simple consequence of natural science. Thereby a firm basis seemed at the same time to have been won for all our representations and for our conduct of life. In place of a mystical or spiritualistic foundation we now get a wholly tangible foundation to build upon in theory and in practice. And it is a doctrine easy to popularize. It is childlike, intuitable, easily accessible, and its presentation gives good occasion to bring forward a mass of natural-scientific facts of general interest. Several of the spokesmen of German materialism were able and independent investigators of nature, such as CARL VOGT and JACOB MOLESCHOTT. Others, like LOUIS BÜCHNER, worked especially by clear, tasteful and enthusiastic presentation. It is in its popularizing tendency that this literature has its greatest merit. A

almindelige Naturkræfter. (Oversigt over det kgl. danske Videnskabernes Selskabs Forhandlinger. 1856), p. 155, 166. — Joule: *Scientific Papers*. I, p. 269, 273. — As regards Mayer, it appears from his address at Innsbruck (*Die Mech. d. Wärme*, p. 357) that he did not assume any break in the series of material phenomena, but assumed a parallelism between the spiritual processes and the brain processes.

wealth of knowledge has been spread abroad by it in wide circles. Büchner's *Kraft und Stoff* is one of the most read popular-scientific books of our century; from 1855 to 1899 nineteen German editions appeared, and it has besides been translated into several other languages. Even if there is something dogmatic about materialism, it has nevertheless done good service by setting its dogmatism against ecclesiastical dogmatism, and thereby awakening reflection upon problems which had been pushed aside at the dissolution of Romantic philosophy. The whole materialist movement in Germany was moreover borne by an idealistic interest in humanity and progress, and Büchner was fully justified in protesting against the confusion of materialism as method and theory with materialism in the sense of a practical direction of life. Materialism can very well acknowledge the value of the highest and noblest ideas and feelings — even if it holds that they, like all spiritual phenomena, are only products of, or forms of, material processes.

When materialism claims to be the simple consequence of the results of natural science, it is of interest to observe — not only that the discoverers of the law of the conservation of energy proceeded throughout from spiritualistic presuppositions, but that the most important collisions in the materialist controversy took place between parties who all stood upon the ground of natural science. This does not necessarily show that materialism is wrong; but it shows how difficult it is to draw the true consequences, and how many different motives come to influence the world-view of individuals. Moleschott's famous book *Der Kreislauf des Lebens* (1852) is directed against LIEBIG's theologizing utterances in his "Chemische Briefe", in which he had particularly attacked an earlier proposition of Moleschott's: "Ohne Phosphor kein Gedanke!" At the same time a controversy broke out between the physiologist RUDOLPH WAGNER in Göttingen and the zoologist CARL VOGT in Geneva. This controversy reached its height when Wagner, at a congress of natural scientists in Göttingen in 1854, declared himself in favour of the assumption of an ethereal soul-substance which moves the brain's fibres as the musician moves the pianoforte, and which is propagated from parents to children by division [!] — an assumption he supported upon the Bible's teaching, but which he admitted he could maintain only by a sharp distinction between faith and knowledge. For when one goes beyond the domain of exact knowledge one has, according to Wagner, only an unprovable faith to hold to — and he adds: "In Sachen des Glaubens liebe ich den schlichten, einfachen Köhlerglauben am meisten." Against this Vogt came forward with a robust polemic: *Köhlerglaube und Wissenschaft* (1855). In a later work, *Vorlesungen über den Menschen* (1863), Vogt gives a more scientific presentation of his standpoint. He lays weight above all upon the brain's being the organ of consciousness, and upon

consciousness standing to the brain as every function stands to the corresponding organ. Although he admits that it cannot be explained *how* consciousness arises in the brain's cells, he holds fast that it is indissolubly bound to them; and he does not give up the teasing proposition he had already set up earlier, that thought stands in the same relation to the brain as bile to the liver or urine to the kidneys — notwithstanding that a function stands in a different relation to its organ than a product does to the place where it is produced.

Moleschott relies, as the title of his work shows, especially upon the doctrine of the conservation of matter. It is for him the great thought which the Encyclopaedists of the eighteenth century had already brought forward, which subsequent natural science has confirmed, and upon which the faith of the future must be built. He feels reverence for the great circulation in nature. The miner fetches phosphate of lime up out of the earth in the sweat of his brow — and perhaps thereby the matter destined for the best head and the highest thoughts passes through his hands: the farmer manures his soil with the phosphate of lime; this becomes a constituent of the wheat that nourishes men's bodies and brains! With matter, life circulates through all parts of the world; with life, thought — and from thought there springs in turn the will to make life better and happier. If we can but supply the organism and the brain with the best substances, thought and will too will attain their highest development. The investigator of nature is our age's Prometheus, and chemistry is the highest science. The social question finds its solution if one can but [sic] find the right distribution of the substances to which the life of thought and of will is bound. — Only the first edition of Moleschott's book has been taken into account here; in later editions it was altered and expanded. Moleschott, who was born in Holland in 1822, was working as a lecturer in Heidelberg when the book appeared; but when his freedom of teaching was interfered with there, he moved to Zürich; later he worked as professor of physiology in Turin and in Rome, where he died in 1893. After his death his autobiography appeared (*Für meine Freunde. Lebenserinnerungen von Jacob Moleschott*. Giessen 1895). It gives a fine picture of the idealistically minded investigator of nature, striving after an all-round culture. For a right understanding of his standpoint the following utterance is of service, which comes after the train of thought of "Kreislauf des Lebens" has been developed: "One-sidedly 'materialistic' this is only for those who can imagine a matter without force or a force without material bearer. I was clearly conscious that one could turn the whole way of representing it round: since all matter is a bearer of force, is endowed with force or penetrated by spirit (*geistesdurchdrungen*), one may just as well call it an idealistic view" (p. 221). Moleschott's standpoint is rather to be

called monism than materialism: “It is a question of a true, indivisible two-in-oneness, and the dispute is not between materialism and spiritualism, but between two-in-oneness (*Zweieinigkeit*) and divided duality (*zwiespältige Anschauung*)” (p. 222).

LOUIS BÜCHNER (1824–1899) does indeed declare that he will not explain what the relation is between spirit and matter, force and substance; he will only maintain that there is a necessary and indissoluble relation between them. But even this is more than can be scientifically proved; and Büchner moreover harbours no doubt that spirit is only a property of matter, force only a property of substance. The impulse to his famous book “Kraft und Stoff” he had received from reading Moleschott’s “Kreislauf des Lebens”, and it is the eternity of matter that is the ultimate foundation in him too. He holds that the conservation of force is really a self-evident consequence of the conservation of matter, and could therefore already have been drawn from Lavoisier’s chemistry. It was only in the fifth edition of his book, however, that he inserted a special chapter on “the immortality of force”, in which the circulation of force is set as a necessary correlate beside the circulation of matter, so that “the two together form, from eternity and to eternity, the sum of phenomena which we call the world”. Yet he constantly holds fast to the assumption that “in matter dwell all the forces of nature and of spirit; matter is the primal ground of all being”. And as regards the spiritual, he regards it — notwithstanding that he has disclaimed all insight into its relation to matter — as a mere product. “As the steam engine produces motion, so the complicated organic composition of force-endowed substances in the animal body produces a total sum of certain effects which, when they are bound together into a unity, are called by us spirit, soul, thought” (Kraft und Stoff. 7. Aufl., p. 130). In a later utterance Büchner goes — without appearing to notice it — a step further. Thus it runs, in an utterance directed polemically against the author of the present work: “The opposition between physical and psychical energy is tenable only so long as one conceives the concepts of body and spirit, or more generally of matter and force, in a dualistic way, whereas for the materialistic-monistic conception they coincide into one and perhaps present only *different sides or forms of expression of one and the same primal ground of all things*. It is precisely the law of the conservation of force that in the problem of the soul leads necessarily to materialistic consequences, as I, supported by physiological-psychological facts and considerations, believe I have established to evidence in several of my writings, and recently again in the fifth of my letters on *Das künftige Leben und die moderne Wissenschaft* (Leipzig 1889). I believe I have there shown that psychical activity *is not, and cannot be, anything other than the radiation, taking place between the cells of the grey cerebral cortex, of a movement initiated*

by *outer impressions*” (from an article in the weekly *Menschthum*. Gotha 1889, No. 46).

It is clear that Büchner in this utterance confuses two different theories: the one, which conceives spirit and matter as forms of expression of one and the same substance, and the other, for which matter is “the primal ground” and psychical activities are therefore consistently *nothing other than* motion (or “radiation of motion”). In one and the same breath he professes both. The tireless champion of the influence of natural-scientific results upon our world-view has not seen this inconsistency, which perhaps in his eyes did not much matter, since his real aim was not to inculcate any one determinate philosophical theory. In his great zeal he has also believed that he had “established to evidence” more than can be established.

With sharper criticism HEINRICH CZOLBE (1819–1873), a physician like Büchner, sought to determine the consequences of the teaching of natural science. In his first works he is a consistent materialist. In *Neue Darstellung des Sensualismus* (1855) and *Die Entstehung des Selbstbewusstseins* (1856), as well as in the paper *Die Elemente der Psychologie vom Standpunkte des Materialismus* (*Zeitschrift für Philosophie*. XXVI), he develops the view that materialism conflicts with natural science so long as the latter assumes the doctrine of the specific energy of the sensory nerves, and thus assumes a difference between the sensations and the outer processes corresponding to them. In opposition to that doctrine Czolbe sets up the assumption that it is essentially one and the same motion which propagates itself from the surrounding world, through sense-organs and nerves, to the brain. At no point does any qualitative change take place; the difference between the different senses is to be explained by the different intensity with which the motion proceeds in the different organs. The unity of consciousness is to be explained by the movements in the brain acquiring a direction that runs back into itself; by making such circular movements possible, the brain is the organ of consciousness. Thus Czolbe would conceive both sensation and self-consciousness, plainly and simply, as movements in space. But — clear and consistent thinker that he is — he becomes aware that if he identifies sensation and self-consciousness with motion without more ado, then one must also be able to reverse the proposition and say that wherever motion of a certain intensity and a certain direction is present, there consciousness is present too, so that the consequence is a universal ensoulment of nature. Thus the consistent carrying through of materialism led beyond it.⁶

⁶The consequence Czolbe draws from materialism was already drawn by Henry More, who (*De anima*. Roterodami 1677, p. 64) sets up the following fundamental proposition: “If sensation can be ascribed to matter, then the motion of one material part, or its reaction upon another material part, or at any rate the continuation of that motion, is one and the same as sensation and apprehension. This proposition ... is confirmed by the voice of my chief opponent, Hobbes.” He then refers to Hobbes’s doctrine of sensation as

Later — in the work *Die Grenzen und der Ursprung der menschlichen Erkenntniss* (1865), and in the exceedingly interesting paper *Die Mathematik als Ideal für alle andere Erkenntniss und das Verhältniss der empirischen Wissenschaften zur Philosophie* (*Zeitschrift für exakte Philosophie*, 1866) — Czolbe saw that it was impossible to explain existence from a single principle, whether one finds that principle with Büchner in matter, or with the speculative philosophers in spirit, or with the theologians in God. We reach an explanation only by proceeding from several elements, calling that an element which we are unable to analyse. Such elements, not reducible to one another, are the material atoms, the organic forces, and the mental elements (which last make up in sum the world-soul). Between these three kinds of element a harmonious co-operation takes place, by which a purposive connection in nature is realized. Not in its origin, but in its direction and tendency, does existence establish itself as a unity. — Thus it appeared that the problem of existence was more intricate than materialism supposed. In his later investigations, which are only partly published, Czolbe approached Spinoza's fundamental thoughts, of which he wished to give an empirical execution. — On the different stages of development of this energetic thinker's philosophizing, reference must here be made to VAIHINGER's paper *Die drei Phasen des Czolbeschen Naturalismus* (*Philosophische Monatshefte* XII).

What lies at the ground in Czolbe from first to last is the demand for clarity and intuitability in all fundamental concepts. He wanted everything mystical and super-sensible away, and sought so far as possible to think all ideas with geometrical clarity. It was more with respect to intuitability than with respect to stringency of proof that he proclaimed mathematics the ideal of science. In his youth he was enthusiastic for Hölderlin's poetry, and had set himself the task of maintaining Greek clarity and joy in life against all mysticism and dualism. Intuitability in the domain of thought was for him closely akin to joy in the natural world. Just as it may require labour to carry through the principle of intuitability in thinking, so it may cost resignation to carry through joy in actual life in practice; but in the theoretical domain as in the practical, Czolbe would combat what he calls "der Blödsinn der Transcendenz". He admitted already in his materialist period that the innermost motive of his philosophy was the subjective need for intuitability and for holding fast to the actual; materialism was to him a postulate — which was later replaced by other postulates. —

ERNST HAECKEL (b. 1834, from 1865 professor of zoology in Jena) is often reckoned to materialism, but has himself determined his standpoint as a monism which leads beyond the opposition between spiritualism and materialism, and which takes as its

a motion. Hobbes, however, denied that consequence, saying expressly that the proposition that sensation is motion cannot be reversed. (*De corpore* XXV, 5).

ground the great pantheistic thought of the unity of nature. For monism, he says, there exists neither spirit nor matter in the usual sense, but only one thing which is both at once. It was in his *Generelle Morphologie* (1862–1866) that he first uttered his general philosophical views. The mental is for him — as for Czolbe — an original element in existence, though it exists in exceedingly different degrees, from the atom-soul and the cell-soul to the souls of the highest organisms. This theory of ensoulment becomes questionable in Haeckel through his often letting the soul's action serve as an explanation of organic movements, instead of seeking a purely natural-scientific explanation. On the other hand he sometimes declares mental phenomena to be mere effects of the most complicated and most unstable carbon compounds, the albumen molecules in nerve tissue. He believes himself to be beyond spiritualism and materialism, and yet one finds in him both propositions that sound spiritualistic and propositions that sound materialistic. He has not arrived at his principles through a sharp thinking-through of the laws of the conservation of matter and of force. It is Darwin's doctrine that set his thinking in motion, and he has sought a highest principle that could be united with the great unity in nature to which the new doctrine pointed. Monism leads, according to Haeckel (*Gen. Morphol.* II, p. 445–451), to the most exalted idea of God. In the great, all-embracing law of causality the Godhead reveals itself as embracing the whole of nature and operating in every natural phenomenon. The usual conception of God, by contrast, is belief in two gods — amphitheism, not monotheism — because it sets the natural causes beside the Godhead.

Haeckel was one of the first German investigators of nature to attach himself to Darwin, and he fought with great ardour for his theory, to which he ascribed a certainty, and gave an extension, that could not always be approved by the theory's critical and level-headed founder. He was untiring in constructing genealogical trees for the species now living, and he had no scruple about assuming a constant arising of organic matter. He has no clear eye for the limits of hypotheses and for the necessity of verification; only thereby is it explained that he can without more ado set Darwin's hypothesis alongside Newton's. He accordingly also reproached Darwin with laying too much weight upon the objections to the theory. The great investigator shook his head at his young adherent's zeal: "Your boldness sometimes makes me tremble", he wrote to him (19 November 1868). — In Haeckel's most recent "philosophical" work (*Die Welträthsel. Gemeinverständliche Studien über monistische Philosophie.* Bonn 1899) the qualities mentioned appear in a still more glaring form than in his earlier utterances. —

The doctrine of the conservation of matter and of force, and the doctrine of the

development of species through natural selection, were — as already remarked — the natural-scientific results which in the period after the middle of the century had above all to set reflection in motion. Having now considered the attempts made by investigators of nature to found a philosophy upon these results, we shall go on to examine the way in which professional philosophers took up their position with respect to them.

3. Idealist Constructions on a Realist Basis

a. Rudolph Hermann Lotze

Idealist philosophy has in Lotze its most important representative in the last half of the century. In his personality and in his development he stands as one who has taken up into himself, in an interesting way, both the ideal motives upon which the philosophy of Romanticism built and that strict carrying through of the mechanical view of nature which towards the middle of the century was working its way forward in science. Lotze is a master of conceptual development — of discussing a thought so that all its nuances come out, of taking up a problem again and again and treating it from different points of view. His ideal was in the last instance the same as the one that floated before the philosophy of Romanticism: to derive all development and all connection in the world from an eternal Idea, which should contain the ultimate ground of all that happens, as well as of the value of what happens. But he saw clearly that such a derivation exceeds what human thought can accomplish, and that when the thinkers of Romanticism nevertheless believed they had given one, it was because in them the poetic-religious tendency involuntarily merged with the philosophical. Lotze therefore separates the poetic-religious element from the speculative. His own fine poetic sense let him understand that their Romantic confusion was not a happy one. And this poetic sense is one with a sense for individual nuances and relations, which speculative philosophy all too often dissipated by its abstractions. Through this sense Lotze's speculative interest hangs together with his realist tendency. What matters for him is first and foremost to apprehend the phenomena in their concrete nature and in their determinate, law-governed connection; it is then the task of philosophical thinking to find out the presuppositions upon which that real connection rests. The poetic, the natural-scientific and the philosophical element in Lotze thus hang closely together, and he was equipped as few have been to treat the task he had set himself, and which the intellectual conditions of the age suggested: to attempt a reconstruction of an idealist philosophy upon a realist basis. It was his conviction that the philosophy of Romanticism had overlooked the real conditions and the mechanical

connection of nature, without which even the most significant ideas are helpless idols; just as it was his conviction that materialism made *that* into the be-all and end-all which was only a — quite certainly necessary — form and frame for the valuable content of existence. His philosophy is essentially an analysis of the very concept of natural mechanism, designed to show that this concept leads with necessity to the assumption of an ideal principle of existence, and that it at any rate does not exclude the assumption that such a principle is the eternal source of all that is good and valuable in the world.

In Lotze's development the different motives and interests that animated him appear clearly. He was born in Bautzen in Saxony on 21 May 1817, in the same region from which Lessing and Fichte come. At the university of Leipzig he studied philosophy, medicine and physics. He was here initiated into the two circles of thought which it was later to be his task to unite, after having worked through them separately. It was the aesthetician and philosopher of religion Chr. Hermann Weisse who was his teacher in philosophy, and Lotze later said that he owed to Weisse not merely many particular influences, but that he was in particular introduced by him into a circle of thoughts which he has since felt no need to give up. Weisse was the most eminent representative of philosophical theism. Through him Lotze's religious-philosophical conception stands in historical connection with Schelling's — and through that with old Jacob Böhme's. Lotze emphasizes, in a retrospect upon his development (*Streitschriften*. 1857, p. 5 ff.), that he appropriated the ideas of speculative philosophy not as a closed doctrinal edifice but as a peculiar form of intellectual culture. The presupposition that the ultimate ground of things could be thought only as spiritual was early laid down in him and never given up. Medicine and physics he studied under E. H. Weber, Volkmann and G. Th. Fechner. Here he came to know the natural-scientific method and conception at first hand. In one and the same year he took the doctorate in philosophy and in medicine, and now came forward as lecturer in both subjects — also after a professorship in philosophy had been conferred upon him in 1842. After working some years in Leipzig he became professor in Göttingen, as Herbart's successor, and here fall his most important works. The year before his death he received a call to Berlin, but there soon succumbed to an illness from which he had long suffered (1881). His life was quiet, devoted to study, thought and academic teaching. With a rare union of many-sidedness and thoroughness he knew how to orient himself in highly different fields — something to which not only his medical and philosophical works but also a whole series of smaller papers and reviews (published after his death in four volumes under the title *Kleine Schriften*) bear witness. From his more rigorous scientific labours he rested in occupation with art and literature: thus he

occasionally translated “Antigone” into Latin verse.

As a medical writer Lotze set himself, as already touched upon, the task of maintaining physiology’s character as a mechanical natural science. He would not have the peculiarity of the organic phenomena explained by appeal to a mysterious vital force, but by exhibiting the determinate, law-governed manner in which the general forces of nature operate within the organism. As everywhere in nature, so here too we must find the explanation in the interaction of real elements. Not by being withdrawn from the mechanical connection of nature, but by the peculiar manner in which connected series of effects arise, does organic life distinguish itself from the inorganic world. In speculative philosophy of nature the idea of the organism had been made the type for the whole conception of the world; that this idea was corrected had therefore to be of general philosophical significance, and one understands that not only Lotze’s medical but also his philosophical interest was satisfied by his labours in this direction (*Allgemeine Pathologie und Therapie als mechanische Naturwissenschaften*. 1842. — *Allgemeine Physiologie des körperlichen Lebens*. 1851). These works even gave some materialists occasion to see a fellow-thinker in Lotze, notwithstanding that he stated definitely that mechanism was for him only a part of his conception of nature, not the whole. It is characteristic of Lotze that he thus divides the problems. His general philosophical views he had developed in his *Metaphysik* (1841). But he by no means felt himself finished when he began his teaching activity, and especially as regards the concluding ideas he constantly referred to future works. In *Medizinische Psychologie oder Physiologie der Seele* (1852) he expressed himself in detail upon the relation between the spiritual and the material, and at the same time gave psychological investigations of lasting interest. A long-cherished plan he carried out in the following period (1856–1864) with the publication of his *Mikrokosmos* (3 vols.). This work, conceived as a counterpart to Humboldt’s “Kosmos” and Herder’s “Ideen”, gives a psychology in close connection with physiology and cultural history, and ends with a development of Lotze’s cosmological and religious-philosophical ideas. For all his interests, for all the different roads his thought and feeling had travelled, he had here found expression. The work is laid out as a popular presentation and has won a considerable circulation. After publishing a spirited *Geschichte der Aesthetik in Deutschland* (1868), Lotze proceeded to give a concluding systematic presentation of his philosophy. But only two parts of it did he succeed in finishing (*Drei Bücher der Logik*. 1874. — *Drei Bücher der Metaphysik*. 1879). The third volume, which was to comprise aesthetics, ethics and philosophy of religion, was not worked out. His philosophy is thus unfinished; it was not granted him to set the crown upon the work and show how

the many threads of thought he had spun could be woven together into unity. — Short presentations of Lotze's teaching in the different philosophical fields are to be had in the dictated notes (*Grundzüge*) from his lectures, published after his death. Among these must especially be singled out *Grundzüge der Logik und Encyclopädie der Philosophie*, which (in its second part) gives a short and clear survey of his whole system. Here we give a characterization of the three most important points in his philosophy.

α. The Mechanical View of Nature

Lotze's thinking has two points of departure: a living feeling for the value of spiritual life, for the fact that the highest thing for us is bound up with spiritual development and its ideals — but at the same time a firm conviction that a system of mechanical causes and laws is necessary to the realization even of the highest ideals. He can, as he says somewhere, believe in ideas that operate, but not in ideas that bewitch. The two points of departure do not, however, stand wholly isolated in him. He seeks, by proceeding from the one, to show the possibility of acknowledging the other. The most characteristic and the most scientific feature of his philosophy is his attempt to establish, through an analysis of the very concept of mechanism, that its ultimate presupposition is the concept of a principle which — as soon as we conceive it in a way intelligible to us — shows itself as the bearer of, and the source of, the highest ideas. Lotze means to take the bull by the horns: by drawing out realism's consequences he will found idealism. Only investigations conducted in realism's spirit can, in his view, bring us nearer to the goal idealism sets itself: the knowledge of the world as the expression of a valuable idea. Romanticism's philosophy had believed it could derive the forms of reality from the highest Idea. That has shown itself to be an insoluble task. We can, by contrast, attempt by way of inference to go back from what is given to its presuppositions. Not a *deduction* but a *reduction* is possible. Thinking must always apply its forms to a given material. Both general culture and the several sciences here operate with a mass of concepts whose origin, meaning and validity they do not investigate more closely. Such concepts are cause and effect, matter and force, end and means, freedom and necessity, matter and spirit. It is philosophy's task to bring unity and connection into the circle of representations by making these concepts — presupposed in practical life and in the several sciences — the object of a peculiar investigation, and by determining the limits of their validity (*Grundzüge der Logik und Encyclopädie der Philosophie*, § 88).

Of these concepts the most important is the one we presuppose in every investigation of reality — every time we make an experiment, every time we seek an explanation: the

concept of a pervasive, all-embracing causal connection. This concept is not grounded upon experience, but is presupposed in all experience. But since the understanding of nature so far won rests upon its reality, it can itself be described as the expression of a fact; and that fact — that the single element in our experience is bound by a law-governed connection to other elements, the fact of mechanical connection — is what philosophy has to think through and to draw all the consequences of. Deduce it, philosophy cannot; but it may perhaps discover what lies in it. A *multiplicity of real elements in mutual interaction* is the foundation upon which the mechanical view of nature builds; and Lotze was, as we have seen, so convinced of that foundation's inevitability that he himself worked with zeal to gain it recognition within physiology, where hitherto vitalism had reigned with its appeal to a single forming and directing "vital force".

But because mechanical connection is an inevitable feature of our conception of the world, it need not be the only, all-determining feature. On the contrary: even if its validity is unlimited, it may perfectly well have a subordinate significance in the world as a whole. (*Die unbeschränkte Gültigkeit des Mechanismus, aber zugleich seine durchaus untergeordnete Bedeutung im Ganzen der Welt! Drei Bücher der Metaphysik*, p. 462.) And a closer analysis of the concept of mechanism leads us to see that this is precisely how the matter stands.

The mechanical view of nature stops — when it is made into a concluding world-view — at a multiplicity of elements (atoms) in mutual interaction. It proclaims a pluralism. But how do the elements stand to the connection in which they stand? Can they have independent subsistence apart from that connection, so that it is a relation indifferent to their essence — or are they not determined right through by the combination in which they stand with the world as a whole? Interaction and connection cannot, after all, float freely above or between the elements, but presuppose an inner unity of them. For if I assume that the two elements *A* and *B* are wholly independent, interaction between them becomes unintelligible. An effect cannot be transferred as a fully finished state from *A* to *B*. What goes on in *A* can acquire significance for what goes on in *B* only if *A* and *B* are not in reality independent, absolutely separated beings, but if their states are states in one and the same, more comprehensive being. A multiplicity of independent beings would make mechanical interaction unintelligible; intelligibility comes only with the assumption of an infinite, all-embracing being, of which the single elements are moments, or points of action. The concept of a force's or an influence's "passing over" from one independent element to another is untenable. Not a *transient* cause (*causa transiens*), only an *immanent* cause (*causa immanens*) is intelligible. The states of one

and the same being can stand to one another as ground and consequent; the states of two mutually independent beings cannot. — Among the numerous presentations Lotze has given of this train of thought, *Logik und Encyclopädie* §§ 99–100 and *Drei Bücher der Metaphysik* §§ 50–81 must be singled out.

By analysis of the concepts of the causal relation and of interaction — the fundamental concepts of the mechanical view of nature — Lotze has thus arrived at the idea of a primal substance, an all-embracing principle. His thinking here moves along the same paths as those which in their day led Spinoza to his concept of substance, and which Kant walked especially in his youthful writings, without however abandoning them in his later works. He is reminiscent of Leibniz too at several points, without there seeming here, any more than in Spinoza's case, to have been any influence through study. When he heard that he was being placed alongside Spinoza and Leibniz, he said (see FALCKENBERG: *Hermann Lotze*. Stuttgart 1901, p. 85) that he did not feel himself to stand in any inner relation to them, though he admitted that there was occasion enough to institute a comparison.

The concept which Romanticism's philosophy would lay at the ground, and from which it would derive everything, becomes for Lotze the last postulate — or, as he also expresses it, the last fact for our thinking. A closer, intuitable development of the universal principle presupposed in the simplest relation of interaction is impossible. We stand here at a *limiting concept* which we can neither do without nor develop. (*Drei Bücher der Metaphysik*, §§ 73 and 246.) It is nevertheless this concept that makes it possible for Lotze to hold fast to what was for him the essential thing in the idealist train of thought into which his teacher Weisse had introduced him. By it he found a connection between the opposed currents of the world of thought. Neither materialism's absolute atoms, nor Leibniz's monads, nor Herbart's reals could for him mark the conclusion of thinking: pluralism must necessarily give place to *monism*. As regards the atoms in particular, Lotze maintains that natural science's interest leads only to assuming elements which are in fact indissoluble for our experience, but that the assumption of a multiplicity of extended elements — even if these are thought of as vanishingly small — cannot be any conclusion for thought. Either one must give up the atom's unity or its extension: within an extended atom every effect will take time and pass from part to part — and these parts will then be more fundamental unities than the whole "atom". In the concluding concept of the atom we must abstract from all extension and think of the atoms as centres of force which — according to the analysis of the concept of mechanism given above — are each of them points of departure for the primal substance's operation. (*Drei Bücher der Metaphysik*, §§ 190–191 and 245; cf. the interesting review of Fechner's

“Atomenlehre” in *Kleine Schriften* III, p. 215–238.)

Lotze’s assertion of the mechanical view of nature, and the consequences he draws from it, make up the most significant part of his philosophy — notwithstanding that it is as a rule not brought forward when Lotze is praised. It is the spiritualistic and theologizing tendency in him that is most frequently emphasized, by admirers and opponents alike. And yet he has earned his greatest merit as a thinker by his analysis of the fundamental concept of the scientific view of nature. It is a defect in him that he does not undertake an epistemological investigation of the limiting concept he arrives at, but lets it stand in the form in which Romanticism’s philosophy and the earlier dogmatism had conceived it. Lotze had in general no understanding of the importance of epistemology.

β. Metaphysical Idealism

In Lotze’s train of thought, so far as we have presented it hitherto, there is still a certain indeterminacy. It might still be asked whether we then have no way at all of forming a closer representation of the elements and of the primal substance. In now bringing forward Lotze’s answer to this question, it must first of all be emphasized that he is entirely clear that any answer to it rests upon analogies, and is determined by motives other than the purely theoretical. The law of analogy, and the need to find our own spiritual being again in existence — which have shown themselves to lie at the ground in all metaphysical idealists, and especially in Leibniz, Berkeley, Schelling, Schopenhauer and Beneke — appear with full consciousness in Lotze.

Lotze is an atomist — but he does not conceive the atoms as material, because extension, as well as all other sensible qualities, must be explained by the interaction of atoms, which therefore cannot themselves have that property. Like life and all qualities of experience, the sensible fact of extension is due to an interplay of force-points, which are in turn to be conceived as initial points of the infinite primal being’s inner operation. Lotze does not regard it as impossible that wholly new beginnings may occur in the course of the world: this does not conflict with universal lawfulness, for the law always expresses only the order in which the states arise, not an outer fate. The number of the elements need not be constant, but may be thought to rise or fall according to the demands of the highest world-purpose. (*Drei Bücher d. M.* §§ 33 and 235.) Nor need the single elements be wholly homogeneous with one another. A certain agreement or commensurability must indeed be present if one world-order is to embrace them all; but a complete likeness is not necessary. A law of nature may very well bind together qualitatively different elements for which we are unable to find any common denominator.

It is in the end not a *logical* but an *aesthetic* necessity that makes existence intelligible to us (*Drei Bücher d. M.* § 59): not a formal consistency in the primal being's operation, but the richness and fullness with which that operation proceeds, is decisive for the constitution of the single elements. At this point Lotze plainly enough passes over from purely theoretical motives to motives of feeling — without either the one or the other enabling him to determine more definitely how great *differences of quality* are compatible with the *commensurability* which the fact of interaction presupposes.⁷

If we would now form a representation at all of the inner nature of the elements, we must conceive them on the analogy of our own spiritual being. The mechanical view of nature really instructs us only about the mutual relation between the elements, not about their inner nature. It concerns the outer circumstances, moves the whole time around the outside of things — is a *cognitio circa rem*. It is perhaps even inclined — like the popular conception of the world — to hold that it is wholly indifferent to things whether we apprehend them or not. In opposition to this Lotze maintains (already in his youthful work *Metaphysik*, 1841, p. 313) that subjectivity is just as much a part of reality as the outer objects are: *Nicht nur was zwischen den Wesen, sondern auch was in ihnen sich zuträgt, ist ein wahres und wirkliches Geschehen*. In his later writings — first in *Medizinische Psychologie* — he goes a step further, maintaining that our own subjective being is the only case in which we know a thing's interior and have a *cognitio rei*, not merely a *cognitio circa rem*: the only way of forming a representation of the interior of other things is then to conceive them on the analogy of ourselves — that is, as feeling beings (if not as representing beings, since feeling is a more original expression of consciousness than representation). The use of this analogy is the only means of thinking things as real beings subsisting for themselves, which are more than mere images for us. We thereby merely make use of the general method that leads the unknown back to the known. (*Drei Bücher d. M.* §§ 96–98.) And the “aesthetic necessity” demands this too, since it cannot be reconciled with an existence in which a great part — perhaps the greatest — should be merely a dark foundation for a mental life extended to only a few regions (see already *Allg. Physiologie* § 129). The elements of existence must therefore be assumed to be ensouled in manifold degrees.

As it stands with the elements of existence, so it stands also with the all-embracing world-ground that makes their interaction intelligible. If we follow the law of analogy and attempt to lead the unknown back to the known, then according to Lotze we must conceive the world-ground as a spiritual being. When inner states are to be bound into

⁷On the indeterminacy in Lotze at this point see Max Wentscher: *Lotzes Gottesbegriff und dessen metaphysische Begründung*. Halle 1893, p. 19–29.

unity and connection, our own inner, spiritual life is the only example we know in which the possibility of such a preservation of unity through the different states is realized — at least approximately realized. Following his teacher C. H. Weisse, Lotze conceives the world-ground as an absolute personality, and he defends this transference of the concept of personality to the absolute being on the ground that only an absolute being can be a personality, because only it has inner independence and originality, while the concept of personality finds only imperfect application to finite beings, which are dependent upon outer conditions. Lotze admits, to be sure, that to a personal life there belong resistance to be overcome, and a capacity to suffer and to receive as well as to act. But when one then asks how an absolute being, subject to no limits, can suffer, Lotze answers (*Grundzüge der Religionsphilosophie* § 34, cf. § 52) that what sets the Godhead's feeling in motion must be "inner products of its own creative imagination"! — It is surely a great question, however, whether such a self-created resistance can mean anything serious — especially as it can be annihilated at whatever moment one pleases. The personal life we know has in any case precisely not self-created and not so easily removable limits to struggle with, and the analogy upon which Lotze relies therefore seems to fail at just the decisive point. And to this it must be added that Lotze — on the most probable interpretation of his obscure and shifting utterances⁸ — differs from Weisse in that he does not hold the form of time applicable to the absolute being. A personal being that does not develop in time — a timeless life, and a timeless suffering and acting: these are concepts that make far too great demands upon our capacity for analogy! — It need only further be remarked that Lotze — as regards both the elements and the world-ground — presupposes that the opposition between spirit and matter is contradictory, so that no

⁸Reinhold Geijer, in his paper *Hermann Lotzes tankar om tid och timlighet i kritisk belysning* (Lund 1886), drew attention to the shifting standpoints Lotze seems to have taken on the question of time's absolute validity. In my paper *Lotze og den svenske Filosofi* (Letterstedtske Tidsskrift 1888) I associated myself with Geijer's interpretation, though I had to conceive the problem itself otherwise than my Swedish colleague. Yet the matter seems to stand otherwise than Geijer, and I with him, had reason to assume. According to what Richard Falckenberg has shown in his paper *Die Entwicklung der Lotzeschen Zeitlehre* (Zeitschrift für Philosophie und philosophische Kritik, vol. 105), the *Grundzüge der Metaphysik* published after Lotze's death date from as early as 1865, and with that an essential support falls away for the assumption that Lotze first denied the reality of time towards the close of his life. When in earlier writings he speaks without qualification of the time-relation as absolutely valid, this is according to Falckenberg a legitimate use of popular representations where the context did not require a more stringent usage. Lotze's definitive conception was the one presented in *Drei Bücher der Metaphysik* (1879) and in *Grundzüge der Religionsphilosophie* (from 1878–79), according to which succession holds for finite beings, while the Godhead is raised above every difference of time. It departs from his earliest presentation (in *Metaphysik*, 1841) in asserting this distinction, which admittedly does not make the whole question clearer — though the *psychological* distinction between succession and abstract time-form may well be defensible. — As little as my Swedish colleague am I able to connect any meaning with the expression "ein zeitloses Geschehen und Wirken" (*Grundzüge der Metaphysik* § 58). My German colleague, by contrast, finds a significant thought in it.

further possibilities could be thought than the two: only on that presupposition is there compelling necessity to conceive the elements and the world-ground as spiritual beings, when one cannot or will not conceive them as material. But there is no logical, only a factual necessity for choosing between spirit and matter in determining the concluding concepts. We in fact know only these two forms of existence; but there would be nothing *unthinkable* in one or more other forms existing. Therefore metaphysical idealism cannot be proved.⁹

Lotze is now also fully conscious that the conclusion of his whole world-view rests upon motives of feeling, not upon strict thinking. The necessity of going back from the multiplicity of the elements to the unity of the world-ground he asserts, indeed, on purely theoretical grounds: “*Gar kein Weltlauf, weder ein harmonischer noch ein unharmonischer, ist mir ohne die Voraussetzung jener Einheit begreiflich, die alles gegenseitige Wirken des Einzelnen erst möglich macht; die Störungen der Dinge durch einander bezeugen die beständige Gegenwart dieses Einen eben so eindringlich, wie das Zusammenstimmen der Kräfte zum Zweck*” (*Drei Bücher d. M.* § 233). But when he leads everything back to a single principle, he knows very well that we cannot construct the world-connection and the elements out of it: we always lack the second premiss! (*Drei Bücher d. M.* § 93.) Only in practical conviction, then, can the thought be held fast that all beings and all events have their ultimate ground in the highest purpose maintained by the world-ground. What that purpose is we do not know — any more than we know how the conflict of forces in nature and the reality of evil in the moral world are compatible with the validity of the world-plan. (*Mikrokosmos*, Buch IX, Kap. 5. — *Drei Bücher d. M.* § 233.) But in an ethical idea Lotze found, at the beginning as at the close of his career, the final conclusion for his thinking: metaphysics goes back for him to ethics; in what *ought* to be lies the ultimate explanation of what *is*. —

If by pantheism one understands the doctrine which maintains an inner relation between God and the world — whether the Godhead is conceived as a personal being or not — then Lotze’s religious-philosophical standpoint is to be described as ethical pantheism. He himself felt closely akin to Fichte; the kinship with Spinoza he would not properly acknowledge. In the use of theological words and turns of phrase he went — especially in *Mikrokosmos* and in the *Grundzüge* of philosophy of religion and of moral philosophy published after his death — further than was fitting and warranted. By his

⁹Compare my *Psykologi*⁴, p. 80; *Psykologiske Undersøgelser*, p. 91 f.; *Filosofiske Problemer* (Universitetsprogram 1902), p. 62–65. — It is interesting that the founder of metaphysical idealism, Leibniz, in his earlier period, like Spinoza, laid great weight upon the duality spirit-matter being purely factual, not logically necessary. *Philos. Schr.*, ed. Gerhardt I, p. 237, 242, 268. Compare Ludw. Stein: *Leibniz und Spinoza*, p. 93 f.

usage (for it is nothing more) it came to look as though he stood nearer to the popular religious circle of ideas than in fact he did. He does, however, occasionally emphasize (especially *Mikrokosmos*, Buch VIII, Kap. 4) the difference between free religiosity and the positive religions.

b. Gustav Theodor Fechner

c. Eduard von Hartmann

4. Criticism and Positivism

a. Friedrich Albert Lange

b. Eugen Dühring